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MILITARY SPECIFICATION

PROPELLANT, M14, FOR USE IN CARTRIDGE 120 mm, TPCSDS-T, M865

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1.0 SCOPE

1.1 Scope. - This specification covers the requirements, quality assurance provisions, and packing of M14 propellant for use in the Cartridge, TPCSDS-T, M865 for the 120 mm, M256 Gun.

1.2 Classification. - The propellant shall be Type I Cylindrical Multi-Perforated grains.

2.0 APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications and standards. - The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

	MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification For
*	MIL-G-155	Graphite, Dry, For Use In Ammunition

STANDARDS

MILITARY

	MIL-STD-286	Propellants, Solid: Sampling, Examination And Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, Attn: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC-1376

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MIL-STD-652	Propellants, Solid, For Cannons, Requirements And Packing

(Unless otherwise indicated, copies of federal and military specifications, and standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings and publications. - The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

ITOP 4-2-504(2)	Safety Testing of Tank Ammunition

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9345265	Container, Steel
12525000	Cartridge, 120 mm, TPCSDS-T, M865
12525001	Projectile, 120 mm, TPCSDS-T, M865

(Copies of drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS

49 CFR 100-199	Department Of Transportation Code Of Federal Regulations Title 49

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFS 100-199 (latest revision)".)

2.2 Order of precedence. - In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 Material. - The propellant shall be M14 Type I, conforming to the requirements of MIL-STD-652 except for the following:

3.1.1 Chemical requirements.

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3.1.1.1 Nitrogen content. - The nitrogen content of the nitrocellulose shall be 13.15 ± 0.05 percent as determined in accordance with 4.5.1.2.

3.1.1.2 Moisture content. - The moisture content shall be 0.60 ± 0.20 percent.

3.1.1.3 Residual solvent. - The residual solvent shall be a maximum of 0.70 percent.

3.1.1.4 Density. - Density shall be recorded for information.

3.1.1.5 Heat of explosion (HOE). - The HOE shall be recorded for information.

3.1.1.6 Relative quickness (R.Q.) and relative force (R.F.). - This test shall be conducted for information only.

3.1.1.7 Hygroscopicity. - The hygroscopicity shall not exceed 1.0 percent at 30 degrees Celsius (C) and 90 percent relative humidity. Data shall be taken for information on the first 5 lots produced.

3.1.1.8 Propellant aging. - The minimum time from the propellant packing date to the proving ground testing shall be 60 days.

* 3.1.1.9 Graphite. - The propellant shall be graphited (glazed) to 0.20 percent maximum as determined by an added basis using Grade III or IV as specified in MIL-G-155.

3.1.2 Physical requirements.

3.1.2.1 Physical properties. - The propellant shall conform to the requirements listed in Table I. Mean variations and standard deviations of the dimensions are expressed as a percentage of the mean dimension.

TABLE I. PHYSICAL PROPERTIES

Length (L)	1.11 - 0.13 centimeters (cm)
Grain diameter (D)	0.56 - 0.04 cm
Perforation diameter (d)	0.05 - 0.01 cm
Average web ($\bar{w}$)	0.10 + 0.01 cm
L/D	1.90 to 2.10
D/d	10.5 to 14.0
Uniformity (L)	6.25% mean variation
Uniformity (D)	4.75% mean variation

3.1.2.2 Bulk density. - The bulk density shall be recorded for information.

3.1.2.3 Clusters. - A cluster is three or more grains stuck together. The propellant shall not contain clusters.

3.1.2.4 Physical appearance score. - The physical appearance score shall be recorded for information.

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3.1.2.5 Appearance and condition. - Finished propellant shall be greenish-gray in color with smooth clean surface, clean cut right angle ends, free of sharp points, dust and foreign material.

3.1.2.6 Weight uniformity. - One hundred propellant grains shall be weighed for information as determined in accordance with 4.5.1.12.

3.1.2.7 Loadability. - The charge weight required to meet the ballistic performance requirements shall be 7.40 kilograms maximum.

3.2 Ballistic assessment requirements. - The propellant when fired in the M256 gun with the M865 projectile shall comply with the following requirements: Performance temperatures shall be from -46 to 63 degrees C.

3.2.1 Velocity. - The propellant shall be assessed to a velocity of 1700 meters per second (mps) at 21 degrees C.

3.2.2 Uniformity of velocity. - The standard deviation of velocity at 21 degrees C shall not exceed 6 mps.

3.2.3 Individual maximum pressure (IMP). - The propellant shall not produce an IMP greater than 6220 bars, nor shall the average pressure plus three standard deviations exceed 6220 bars at any temperature.

3.2.4 Negative differential pressure. - The absolute value of the negative differential pressure shall not exceed 345 bars.

3.2.5 Burning residue, smoldering residue, or flareback.

3.2.5.1 Burning or smoldering residue. - The propulsion ignition system shall function such that no burning or smoldering residue shall occur after breech opening in the gun chamber or breech area as observed on an infrared enhanced video recording system when tested in accordance with paragraph 4.5.2.2.2.

3.2.5.2 Flareback. - Flareback of any duration after case base extraction shall be cause for referral (see 4.5.2.2.2). Flareback is the ignition and combustion of unconsumed residual gases with flames extending aft of the breech ring.

3.4 Preproduction inspection. - This specification contains provisions for preproduction inspection. Requirements for the submission of preproduction inspection samples shall be as specified by the Contracting Officer.

3.5 Workmanship. - The best commercial practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. Unless otherwise specified, no reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification.

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4.0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - Preproduction Sample Inspection and Quality Conformance Inspection shall be conducted on this item.

4.3 Preproduction sample (pilot lots). - The preproduction sample is intended to provide the necessary information to establish the propellant granulation needed to meet the required ballistic parameters.

4.3.1 Submission. - The contractor shall select up to three propellant granulations which, from experience, are expected to bracket the requirements of this specification. The contractor shall submit a preproduction sample as designated by the Contracting Officer for evaluation in accordance with 4.3.2, consisting of up to three pilot lots of a maximum of 400 kilograms each, representing the candidate granulations.

4.3.2 Inspections to be performed. - Each of the granulations comprising the preproduction sample and submitted in accordance with 4.3.1 may be subjected by the Government to any or all of the requirements of the applicable drawings and specifications.

4.3.3 Rejection. - Government Engineering and Product Assurance personnel and the Contractor shall review the pilot lot test data and shall select a granulation size which best fits the requirements of this specification. The Technical Agency shall notify the Contracting Officer of the selected granulation. The Contractor shall request approval from the Contracting Officer to produce the selected granulation. If none of the pilot lots submitted meet the requirements of this specification, a new preproduction sample shall be submitted in accordance with 4.3.1. All future production lots should be of the same basic granulation.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing.

PRECAUTION: This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

* 4.4.2.1 Sampling for laboratory testing. - Each lot of propellant

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shall be sampled and tested for chemical and physical properties in accordance with MIL-STD-652. The graphite glaze content, as measured on an added basis, shall be determined in accordance with Method 309.1.2 or 308.1.4 of MIL-STD-286.

4.4.2.1.1 Samples for grain physical appearance score. - A sample of 100 randomly selected grains shall be visually inspected from each inspection sublot of propellant after cutting. The sample will be taken on a routine schedule in order that the average of the samples will be representative of the lot. The grain physical appearance score shall be recorded for information.

4.4.2.2 Sampling for ballistic testing. - A random sample of ten (10) containers or more shall be taken from each lot of propellant sufficient to meet the quantities required for performance tests described in section 4.5.2. After ballistic samples are removed, the containers shall be resealed and held pending ballistic testing. Upon successful completion of ballistic testing, the containers may be returned to their respective lots. To evaluate performance, a minimum of eight (8) valid data points are required per temperature phase, for pressure and velocity. Fewer data points than this require the lot to be referred.

4.4.2.3 Charge establishment. - If the lot does not meet the applicable requirements, testing shall cease and the Test Director or Contractor shall immediately notify the procuring activity.

4.4.2.4 Velocity uniformity. - If the velocity standard deviation of the rounds tested in accordance with 4.5.2.2 exceed the applicable requirements, apply referral criteria in accordance with 4.5.2.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.5 Individual maximum pressure (IMP).

4.4.2.5.1 If any individual pressure observed during the uniformity series, when corrected to the recommended charge weight, exceeds the applicable requirement, the lot shall be rejected.

4.4.2.5.2 If the average pressure plus three standard deviations exceeds the applicable requirement, apply referral criteria in accordance with 4.5.2.4. If the referral criteria cited do not apply, the lot shall be rejected.

4.4.2.6 Negative differential pressure. - If the negative differential pressure of the rounds tested in accordance with 4.5.2.2.1 fails the applicable requirement, the lot shall be referred to the Procuring Activity and shall be evaluated in accordance with ITOP 4-2-504(2).

4.5 Test methods and procedures.

4.5.1 Manufacturing assessment.

4.5.1.1 Grain physical appearance score. - The complete outer

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surface including both ends of the grains will be visually inspected with an unaided eye for the following defects. The number of defects (T) found for each category will be recorded using Table II.

TABLE II. GRAIN DEFECTS

Priority	Category	No. Grains
1	Cracks	III
2	Pinholes	IV
3	Perforation a) distorted b) closed c) partial	II
4	Bad Cuts a) shorter (3/4) b) longer (1 1/4) c) cuts in grain	V
5	Surface	I
6	None	-

NOTE: One is the highest priority of a group with six being the lowest priority and a is the highest priority within a group. Any grain with multiple defects shall be recorded once in the highest priority defect category.

4.5.1.1.1 Category I (surface). - A defect that will appreciably increase the surface area of grain as follows:

4.5.1.1.1.1 Die imperfections. - Grains marked by die imperfections causing longitudinal breaks in the surface of the grain.

4.5.1.1.1.2 Mashed or crushed grain. - Mashed or crushed grains in which the perforation is closed or the grain is cracked.

4.5.1.1.1.3 Blisters. - Blisters or enclosed raised spots on the grain.

4.5.1.1.1.4 Foreign material. - Foreign material, e.g., wood, metal specks, brush hairs, grease, etc.

4.5.1.1.1.5 Indentations. - Cutting machine, roller indentations pushed into the grain surface.

4.5.1.1.1.6 Barbed surface. - Barbed surface, typically slivers or blobs will protrude from the grain surface.

4.5.1.1.2 Category II (perforation). - A defect caused by improper location of the perforation, as follows:

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4.5.1.1.2.1 Closed perforation. - Grains with any one of the perforations closed.

4.5.1.1.2.2 Partially closed perforations. - Grains with any one of the perforations appreciably different from the others (perforations in a grain must not vary in size by more than a 1:.75 ratio nor be more than 25% closed).

4.5.1.1.2.3 Off-center perforations or distorted web. - Grains in which the distance between any 2 adjacent perforations is approximately 1/3 less than the distance between any other 2 adjacent perforations or in which the distance from an outer perforation to the surface of the grain is 1/3 less than the similar distance from any other perforation.

4.5.1.1.3 Category III (cracks). - The surface of a grain shall have no breaks, ruptures, nor cuts.

4.5.1.1.4 Category IV (pinholes). - Grains shall have no visible small holes in the surface (use of a magnifying glass is not permitted).

4.5.1.1.5 Category V (bad cuts). - A defect caused by cutting machine action, as follows:

4.5.1.1.5.1 Short grains. - Grains shorter than approximately 3/4 of the specified length of propellant being inspected.

4.5.1.1.5.2 Long grains. - Grains longer than approximately 1 1/4 of the specified length of propellant being inspected.

4.5.1.1.5.3 Cuts. - Grains which contain cuts in the grain body made by machine blades.

4.5.1.1.6 Calculation of score. - Calculate a physical appearance score for the sublot and lot. The lot score shall be based on the average of all accepted sublots used in the formation of the lot. Report an average and standard deviation for the lot. The sublot score shall be calculated as follows:

$$\text{Score} = 100 = (.5T_I + T_{II} + 3T_{III} + 2T_{IV} + T_V)$$

Where:

T_I = No. of Category I defects observed.

T_{II} = No. of Category II defects observed.

T_{III} = No. of Category III defects observed.

T_{IV} = No. of Category IV defects observed.

T_V = No. of Category V defects observed.

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4.5.1.2 Nitrogen content. - The nitrogen content of the nitrocellulose shall be determined in accordance with method T209.11 of MIL-STD-286.

4.5.1.3 Moisture content. - The moisture content shall be determined in accordance with Method 101.5, 102.1.3 or 103.5.1 of MIL-STD-286.

4.5.1.4 Residual solvent. - The residual solvent content shall be determined in accordance with Method 103.4.1 or 103.5.1 of MIL-STD-286.

4.5.1.5 Density. - The absolute density shall be measured in accordance with Method 510.1.1 except the pycnometer shall be large enough to hold the required sample and a mercury density adjustment for room temperature shall be utilized.

4.5.1.6 HOE. - The HOE shall be determined in accordance with Method 802.1 of MIL-STD-286.

4.5.1.7 R.Q. and R.F.. - R.Q. and R.F. values will be determined by firing against a known standard. Standards will be adopted, established, maintained, and evaluated by the processing facility with concurrence of the direct purchasing agency.

4.5.1.8 Hygroscopicity. - The propellant sample shall be subjected to the hygroscopicity test in accordance with method 503.1.3 of MIL-STD-286.

4.5.1.9 Bulk density. - The bulk density shall be determined in accordance with Method 502.2.1 of MIL-STD-286 except a sample size of approximately 7.0 liters shall be used and 1000 milliliter graduate cylinder shall be replaced with a cylindrical container of a known volume.

4.5.1.10 Clusters. - Two 10 pound samples per shift of production at the final screen prior to pack shall be taken and visually examined to verify compliance with the applicable requirements. Any three or more grain clusters requires an examination of the screen and a rescreen of all propellant that the sample represents. The screen for this process shall be a vibratory screen with openings of 0.95 by 1.91 cm maximum.

4.5.1.11 Appearance and condition. - The propellant sample shall be visually examined to verify compliance with applicable requirements. Failure to meet the requirements shall result in rejection of the lot.

4.5.1.12 Weight uniformity. - One hundred propellant grains shall be randomly selected from the lot and weighed.

4.5.2 Proving ground assessment. - These ballistic tests shall be conducted at a proving ground (designated by the Contracting Officer) in accordance with the applicable proving ground acceptance test procedures. All cartridges (calibration and test) shall be temperature conditioned in a large conditioning box, with adequate air circulation

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to assure uniform conditioning. Assurance shall be made that, when fired, the rounds are at the condition temperature for a minimum of 24 hours. The following tolerances apply to all temperature conditioning:

21 ± 1.5 degrees C
-46 + 3.0 degrees C
+63 - 3.0 degrees C

The cartridges shall be loaded and inspected by an Army Ammunition Plant LAP facility or similar manufacturers of cartridge lots in accordance with drawing 12525000 and DOD-C-64010 or government approved standard operating procedure for ballistic acceptance testing using projectiles fabricated in accordance with drawing 12525001 and shipped to proving ground for ballistic testing. The weight of the lightest projectile used in the firings shall not differ by more than 20 grams from the heaviest projectile used in the firings. These tests should be conducted in the M256 gun with an operative bore evacuator.

4.5.2.1 Charge establishment. - Three groups of three rounds each loaded with selected charge weights of propellant (weights selected should bracket expected service charge weight) shall be conditioned and fired at 21 degrees C. Three calibration rounds shall also be conditioned and fired at 21 degrees C, prior to the test rounds. From the data obtained, a charge weight shall be calculated which will yield the muzzle velocity prescribed in Section 3. This calculated charge weight shall then be loaded into rounds for the uniformity series (see 4.5.2.2).

4.5.2.2 Uniformity. - Thirty rounds shall be assembled with the calculated charge weight as determined in 4.5.2.1. Ten rounds shall be conditioned to 21 degrees C as indicated by a thermocoupled control round. These ten rounds shall be fired alternately with calibration rounds and shall be corrected to standard weapon/firing conditions (i.e., the difference between calibration round performance under test conditions and standard conditions). The mean and standard deviations of the velocities and pressure shall be calculated using corrected results. The charge weight shall be adjusted to obtain the velocity as listed in Section 3 using the slopes of the velocity versus charge weight curve established in 4.5.2.1. The resultant shall be the recommended charge weight. The pressure will also be corrected for any charge weight adjustment. Ten rounds shall be conditioned and fired at -46 degrees C and ten rounds conditioned and fired at 63 degrees C.

4.5.2.2.1 Differential pressure and ignition delay. - Two external pressure transducers mounted 8.39 (rear) cm and 48.26 (forward) cm from the rear face of the tube shall be used for measuring chamber pressure as a function of time during the -46 degrees C, 21 degrees C and 63 degrees C uniformity phases. Differential pressures shall be determined by subtracting the forward gage pressure from the respective rear gage pressure and recording the maximum initial negative difference. Any trace which does not contain any amplitude in the negative direction shall have a negative differential pressure of zero. Ignition delay

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shall be recorded on all differential pressure firings as the time required to obtain 10 percent of the maximum pressure on the rear gage from initiation of the primer.

4.5.2.2.2 Burning residue, smoldering residue, or flareback. - Three cameras, a color video camera, an infrared (IR) enhanced video camera and an infrared video camera (Inframetrics model #600 or equivalent), shall be used to observe burning residue, smoldering residue, or flareback. The color video camera and the infrared video camera shall be used for information only. Acceptance shall be determined by the infrared (IR) enhanced video camera. The cameras shall be positioned such that a full view down the tube is recorded. External light effects shall be minimized to make burning residue, smoldering residue, or flareback occurrences easily detectable. Videotapes of each firing shall include the test identification with propellant lot number, individual round identification, and a clearly legible timer (clock) that records at least to tenths of a second. Evidence of burning or smoldering residue after breech opening shall be cause for referral of the lot to the technical agency for determination of acceptability. Flareback of any duration, as observed on the IR enhanced camera, shall be cause for referral of the lot to the technical agency for determination of acceptability. After each propellant lot acceptance test, original videotapes from all three cameras shall be forwarded to: Commander, ARDEC, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000. When there is evidence of burning residue, smoldering residue, or flareback of any duration, the following shall be provided along with the tapes for technical evaluations:

a. Case base assembly(ies) from cartridges exhibiting burning residue, smoldering residue, or flareback.

b. Any residue remaining in an ejected case base or in the chamber shall be carefully removed and bagged in a clean sealed plastic bag that is marked with the test round number. Photographs shall be taken of the area where a residue indication was observed.

c. Evidence of slow or partial case base extraction or failure of breech to open on counter recoil.

d. Evidence of a damaged, improperly assembled or otherwise improperly operating bore evacuator.

e. Serial numbers of: Tank/mount, recoil, and tube.

f. Evidence of clogged or blocked primer vent holes.

g. Evidence of flaming/sparking primer components in the chamber or case base upon extraction.

h. Magnitude of head wind.

Any indication of burning residue, smoldering residue or flareback on control rounds, or conditioning rounds that are from a different lot than the one being tested shall be reported on the firing record. If conditioning rounds are from the lot that is under test, they shall be

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included with test rounds in the evaluation of burning residue, smoldering residue or flareback.

4.5.2.3 Retest criteria. - All original and retest data shall be submitted to U.S. AMCCOM, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000. A lot shall be retested whenever any of the following conditions exist or whenever the proving ground deems it necessary:

a. If for any reason, the proving ground considers that test conditions have detrimentally affected test results, additional rounds as required shall be tested.

b. If the final adjustment in charge weight exceeds 56 grams of the charge weight used in the uniformity firings, retest for charge establishment in accordance with 4.5.2.1.

4.5.2.4 Referral criteria. - A test lot shall be referred to the Procuring Activity for disposition whenever any of the following conditions exist or whenever the proving ground deems it necessary:

a. Calibration round fired fails to comply with Section 3 of the specification.

b. Calibration round correction applied exceeds 8 mps.

c. Number of test or calibration results utilized in calculations is less than specified.

d. Any unusual occurrences during ballistic testing, such as excessive delays between rounds, poor projectile flights, range instrumentation errors, etc. Each such event shall be reported in detail on the applicable firing record.

e. Test lot fails the muzzle velocity standard deviation or the mean peak chamber pressure requirement at 21 degrees C or 63 degrees C, due to a test value identified as an outlier. The following test shall be performed to determine if a single test result is an outlier. With a sample size of 10 at a significant level of 0.055, the critical value (TC) is 0.477. Calculate a test value (Tt) as follows:

For Low Outlier

$$T_t = \frac{(X_2 - X_1)}{X_9 - X_1}$$

For High Outlier

$$T_t = \frac{(X_{10} - X_9)}{X_{10} - X_2}$$

Where:

X_1 = Test result being tested as low outlier.

X_2 = Test result adjacent to X_1, after all the test results are ordered from smallest to largest.

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X_9 = Test result adjacent to the highest test result.

X_{10} = Test result being tested as high outlier.

X_1 or X_{10} = An outlier if T_t is greater than T_C.

5.0 PACKAGING

5.1 Packing and marking. - The propellant shall be packaged in the metal drum fabricated in accordance with drawing 9345265.

6.0 NOTES

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. - This propellant is designed for use in the M865 cartridge.

6.2 Acquisition requirements. - Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

6.3 Surveillance sample. - Samples shall be forwarded to U.S. Army, AMCCOM, ARDEC, ATTN: SMCAR-AEE-WE, Picatinny Arsenal, NJ 07806-5000, for the propellant surveillance test program in accordance with MIL-STD-652.

6.4 Reference sample. - Whenever a propellant lot is selected for reference use, a 2.3 kilogram sample shall be forwarded to U.S. Army AMCCOM, ARDEC, ATTN: SMCAR-AEE-BP, Picatinny Arsenal, NJ 07806-5000, for use in evaluating the propellant stockpile.

6.5 Standard weapon conditions. - Standard weapons conditions refer to the weapon model and conditions of the test at the time the calibration series was fired.

6.6 Submission of inspection designs for approval (see MIL-A-48078). - Submit equipment designs, as required, to Commander, ARDEC, ATTN: AMSMC-QAR-I(D), Picatinny Arsenal, NJ 07806-5000.

6.7 Equivalent test methods. - The test methods given in the specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed

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method should be submitted through the Contracting Officer to Commander, ARDEC, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision, and drawings of any special equipment required.

6.8 The Technical Agency shall be Commander, ARDEC, ATTN: SMCAR-CCH-V , Picatinny Arsenal, NJ 07806-5000.

6.9 Subject term (key word) listing.

Ammunition
Cannon
Physical appearance score
Propellant
Tank

Custodian:
Army (AR)

Preparing activity:
Army (AR)

Project number:
1376-A417

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-P-70956A	910805

3. DOCUMENT TITLE

PROPELLANT, M14, FOR USE IN CARTRIDGE 120MM, TPCSDS-T, M865

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER, U.S. ARMY ARMAMENT RESEARCH DEVELOPMENT AND ENGINEERING CENTER	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) ATTN: AMSMC-QAA(D) PICATINNY ARSENAL, NEW JERSEY 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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